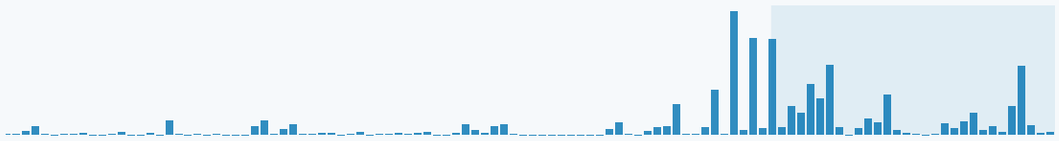
# GammaConv-Net

Prior-guided rainfall response + learned temporal features

## Rainfall history



Normalize / take the latest 30 daily observations

$P_{t-30}, \ldots, P_{t-1}$

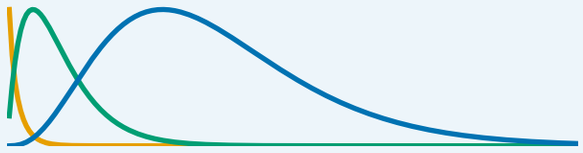
## Observed flow history

Normalize / take the latest 30 daily observations

$Q_{t-30}, \ldots, Q_{t-1}$

## 01 Rainfall response

Learnable Gamma kernels

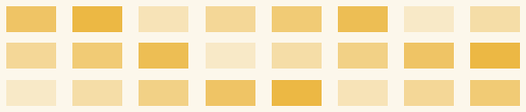


Causal convolution

Rainfall-response feature $q_{base}$

## 02 Rainfall features

Linear layer + ReLU



Mean over the input window

Rainfall feature vector

## 03 Flow memory

LSTM encoder

Last hidden state

Temporal feature vector

**Concatenate features**

30 observed days

1-day-ahead prediction

**MLP fusion + Softplus output**

Inverse flow scaling

**Next-day flow** $\hat{Q}_t$

The dual model also fuses allocation weights. Kernel shapes are schematic.